DEEP LEARNING COURSE · UNIT 1

Introduction to Deep Learning

Neural Networks · ANN · History of AI · Why Go Deep · Applications · Challenges

AI & Machine Learning Department | 10 Hours

COURSE MAP

What We Will Cover in Unit 1

1

Neural Networks & ANN

What neurons, layers and networks are

2

Why Deep Learning

How DL differs from classical ML

3

Brief History of AI

From Turing (1950) to Generative AI

4

Successes of the Last Decade

AlexNet, AlphaGo, ChatGPT and more

5

Reasons to Go Deep

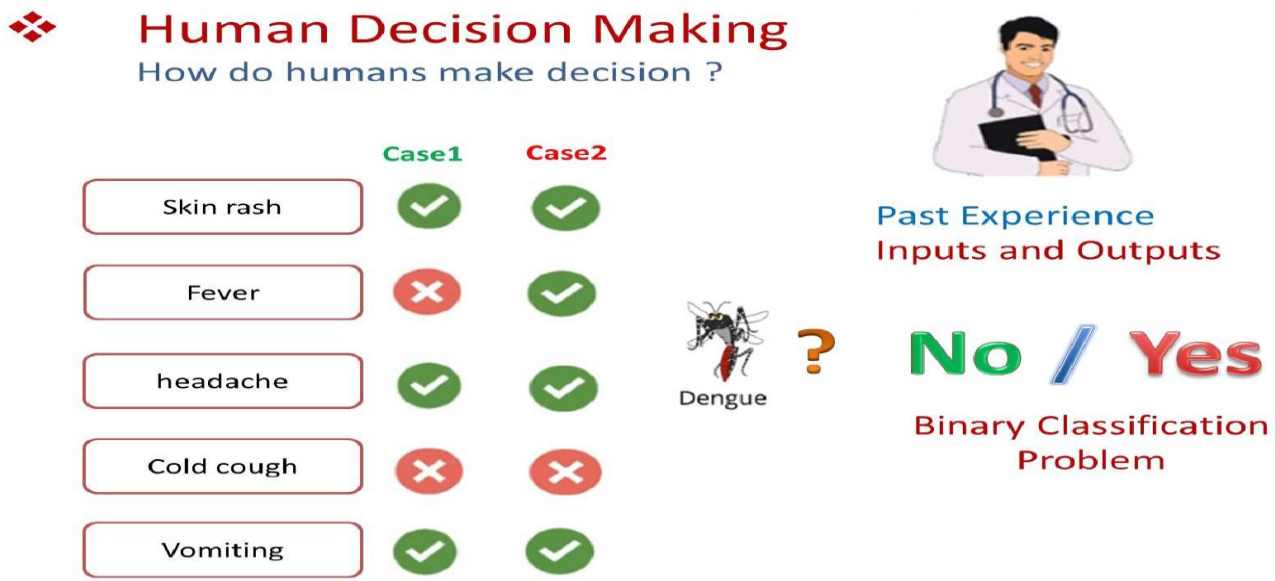
Data, compute, algorithms, features

6

Applications & Challenges

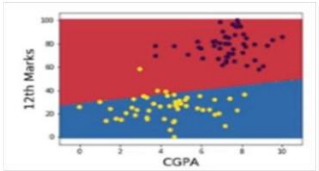
Where DL works, and where it struggles

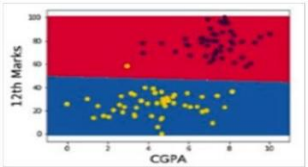
How Do Humans Make Decisions?



From Rule-Based Systems to Data-Driven Models

Rule-based System to Data-driven







Function

$y = mx + c$

$\hat{y} = f(x_1, x_2)$

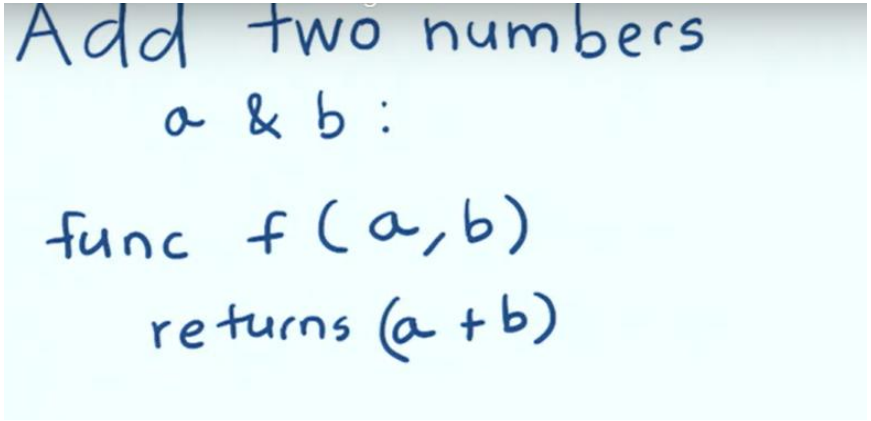
10th marks	12th marks	graduate	CGPA	projects	.....
92	82	yes	9.2	3	.....
83	75.2	yes	7.2	1	.....
75	70	no	6.4	0	.....
96	95	yes	9.5	5	.....
90	89	yes	8.8	2	.....
78	82	yes	7.6	0	.....
86	88	yes	8.4	1	.....



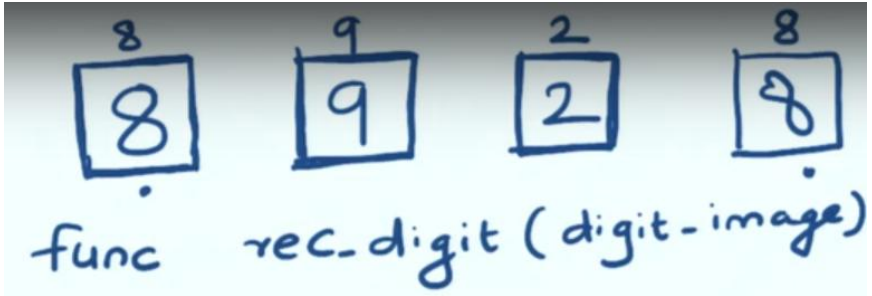


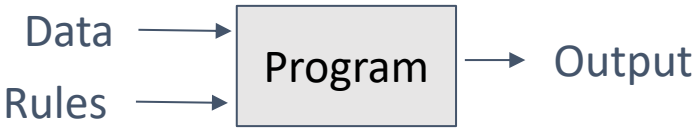
Hire /
Don't Hire

Machine & Deep Learning



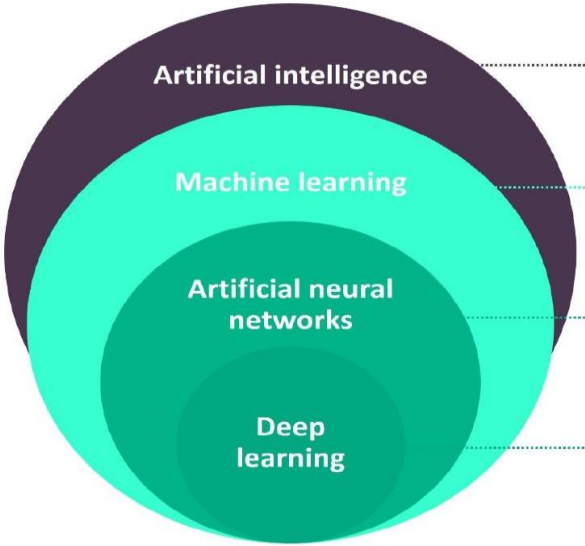
Handwritten digit recognition from images







Where ANN Fits: AI → ML → ANN → Deep Learning



Artificial intelligence (AI)
Any techniques that enable machines to solve a task in a way like humans do

Machine learning (ML)
Algorithms that allow computers to learn from examples without being explicitly programmed

Artificial neural networks (ANN)
Brain-inspired machine learning models

Deep learning (DL)
A subset of ML which uses deep artificial neural networks as models and automatically builds a hierarchy of data representations

02

Why Deep Learning?

Automatic, hierarchical feature learning from raw data

Why “Deep”? Hierarchical Feature Learning

In classical ML, a human engineer manually decides what features matter (edges, colour histograms, etc). In Deep Learning, each layer automatically learns increasingly abstract features from the raw data itself.



Each successive hidden layer composes the patterns detected by the previous layer into something more abstract — this is what “deep” refers to: many stacked layers of representation.

Machine Learning vs Deep Learning

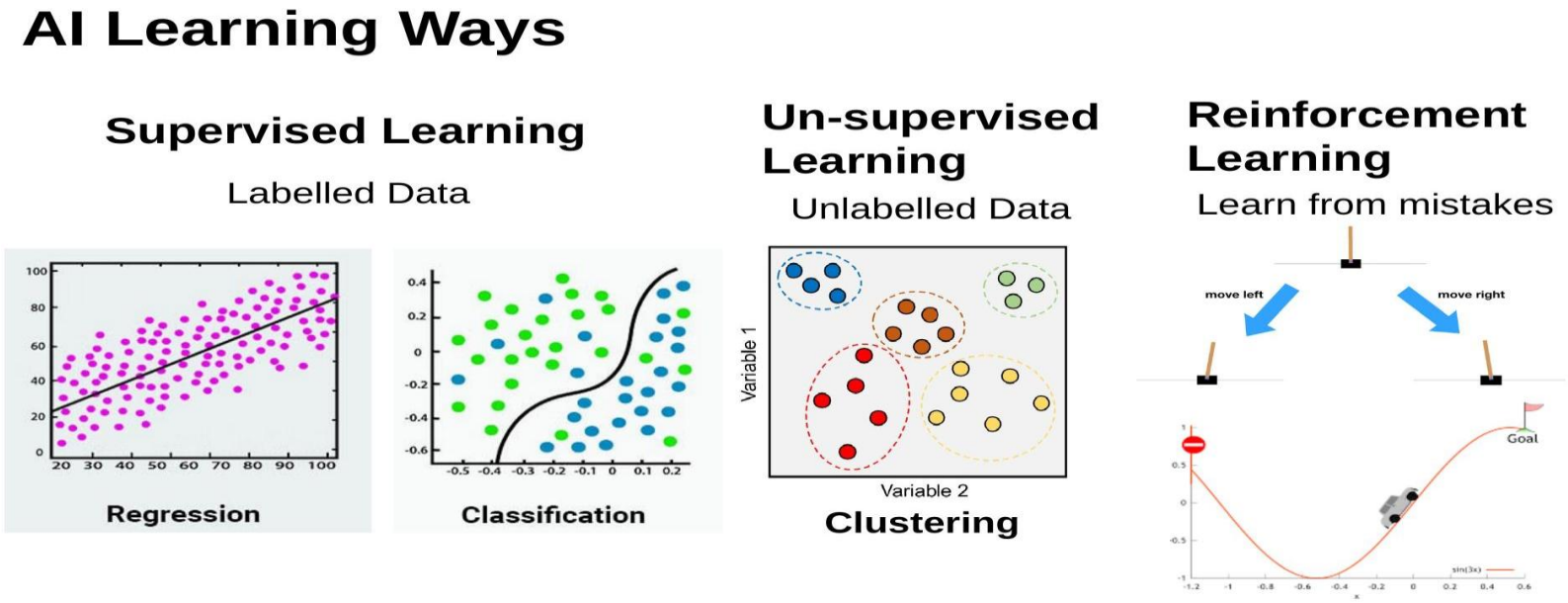
Machine Learning

- Learns patterns from data
- Example algorithms: Decision Trees, SVM, k-NN
- Limitation: Needs **feature engineering** (manual selection of image features)

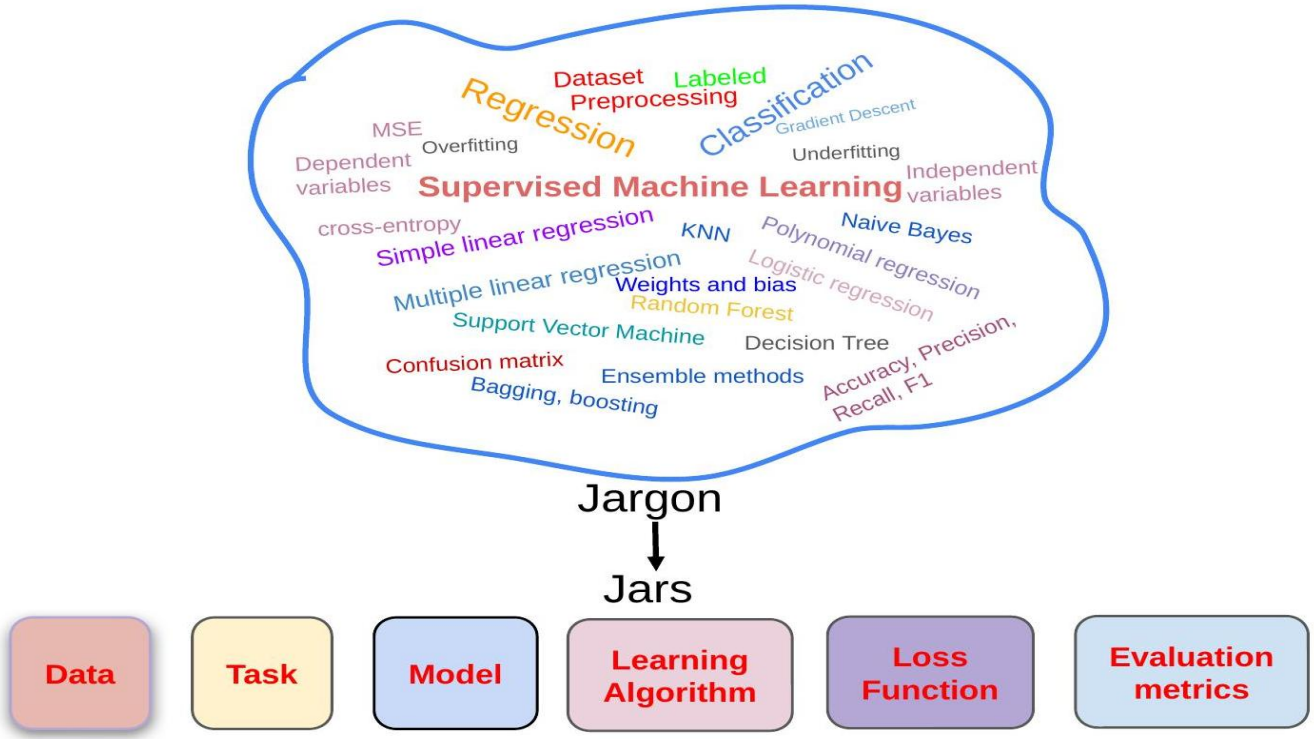
Deep Learning

- Learns features and patterns **automatically** from raw data
- Powered by **neural networks**, especially **Convolutional Neural Networks (CNNs)** for image data
- Requires large datasets and high computational power

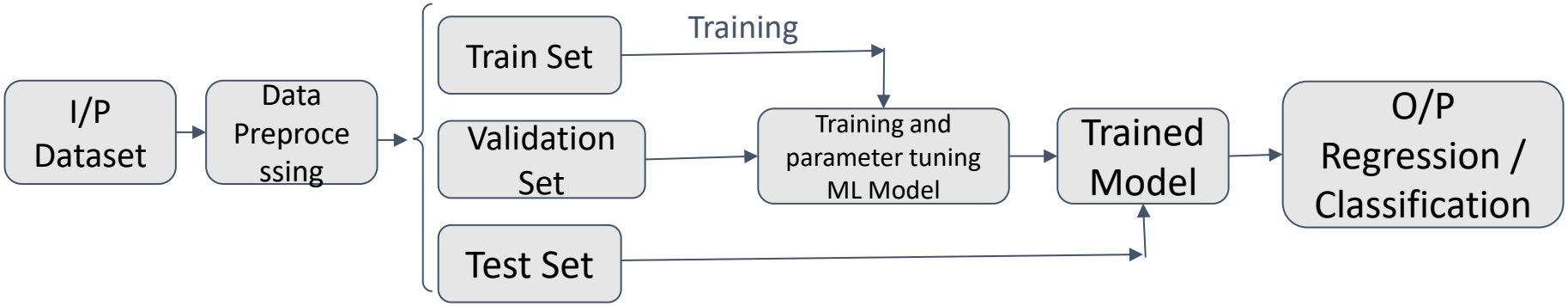
Recap: The Three Ways Machines Learn



Key Terminology Recap



General block diagram for ML/DL



Train – Validation – Test Split



Illustrative split — exact ratios (e.g. 80/20 train-test, with 20% of train held out for validation) vary by project.

Training Set

Used to fit the model — the network's weights and biases are updated directly from this data every epoch.

Validation Set

Checked after every epoch, but never trained on. Used to tune hyperparameters and monitor for overfitting (e.g. via EarlyStopping).

Test Set

Held out completely until the very end. Used exactly once, to report an unbiased estimate of real-world performance.

Key idea: the test set must remain completely unseen during training and tuning — otherwise, performance metrics become overly optimistic and don't reflect real-world generalization.

01

Neural Networks & ANN

Understanding the building blocks of Deep Learning

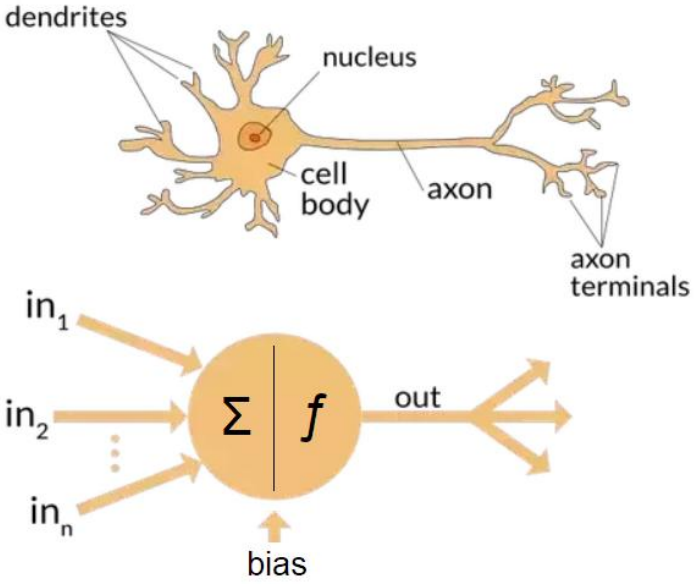
What Is a Neural Network?

A Neural Network is a computing system inspired by how biological neuron process information.

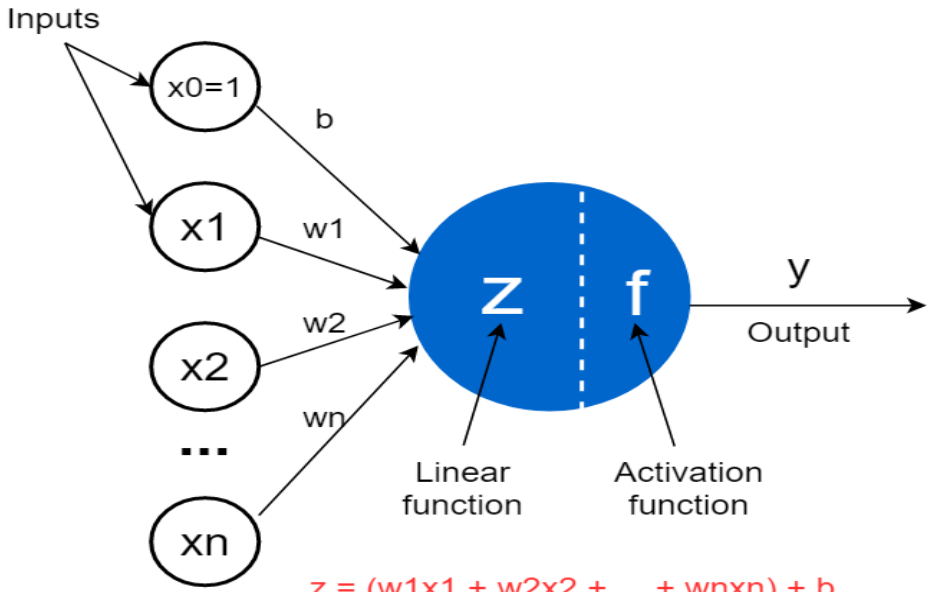
Biological neurons:

- Receive signals through dendrites
- Process the signal in the cell body
- Fire an output via the axon if a threshold is crossed

Biological neuron	Artificial neuron
Dendrites	Inputs
Soma (cell body)	Weighted sum (Σ)
Firing threshold	Activation function (f)
Axon output	Output (y)
Synaptic strength	Weights (w)



The Artificial Neuron (Perceptron)



$$z = (w_1x_1 + w_2x_2 + \ldots + w_nx_n) + b$$

$$y = f(z)$$

The Artificial Neuron (Perceptron)

Mathematical Model

$$z = (w1 \cdot x1 + w2 \cdot x2 + \ldots + wn \cdot xn) + b$$

$$y = f(z)$$

- xi — input features
- wi — weight (importance of input i)
- b — bias (shifts the decision boundary)
- f — activation function (Sigmoid, ReLU, Tanh)
- y — neuron's output

Why Activation Functions Matter

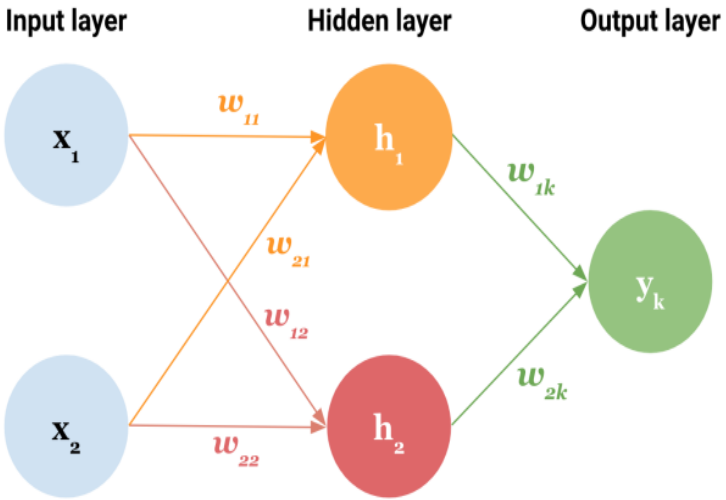
Without an activation function, stacking layers is mathematically equivalent to a single linear layer — the network could only learn straight-line relationships.

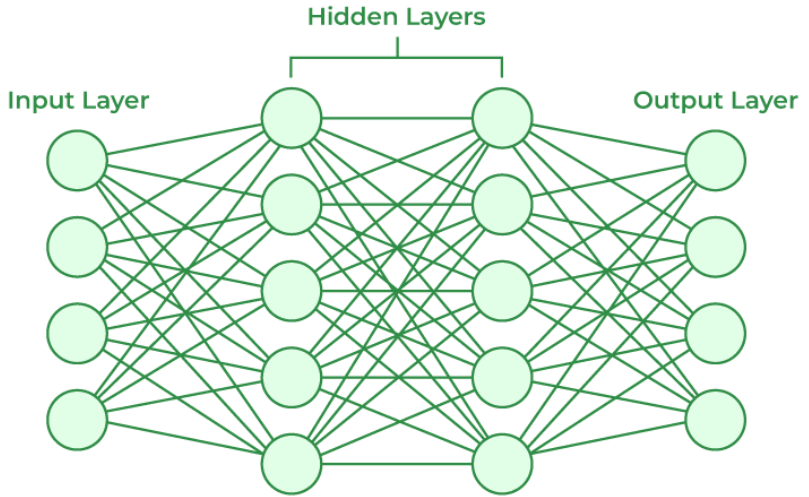
Non-linear activations let networks model curved, complex decision boundaries — essential for images, speech and language.

Common choices: Sigmoid (0–1 probabilities), Tanh (−1 to 1), ReLU (fast, default for hidden layers), Softmax (multi-class output).

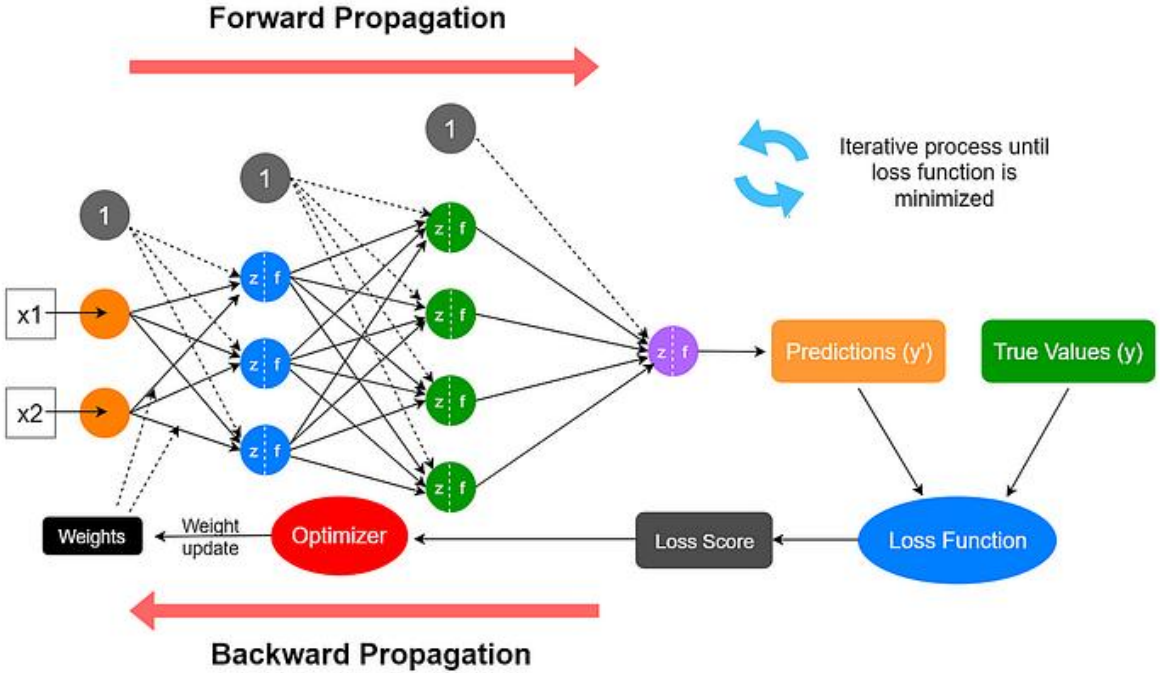
From a Single Neuron to a Network (ANN)

An Artificial Neural Network (ANN) connects many neurons in layers. Each layer transforms the data a little more, building up the ability to represent complex patterns.





How Does an ANN Learn?



How Does an ANN Learn?

1

Forward Pass

Input data flows through the network; each neuron computes a weighted sum + activation, producing a prediction.

2

Loss Calculation

A loss function measures how far the prediction is from the true label (e.g., cross-entropy, MSE).

3

Backpropagation

The error is propagated backward, computing how much each weight contributed to the mistake.

4

Weight Update

Gradient Descent nudges every weight slightly to reduce the loss; the cycle repeats for many epochs.

03

Brief History of AI

From Alan Turing's 1950 question to today's Generative AI

The Early Decades: Birth & Winters (1950–1993)

1950

Alan Turing asks “Can machines think?” and proposes the Turing Test.

1956

John McCarthy coins the term “Artificial Intelligence” at the Dartmouth Conference.

1960s–70s

Rule-based, symbolic AI systems gain popularity.

1974–80

First AI Winter — funding declines as computing power and results fall short of expectations.

1980–87

Expert Systems flourish (e.g., medical diagnosis systems).

1987–93

Second AI Winter.

The Modern Era: From Deep Blue to Generative AI

1997

IBM Deep Blue defeats world chess champion Garry Kasparov.

2012

AlexNet wins ImageNet — the Deep Learning revolution begins.

2016

DeepMind's AlphaGo defeats Go champion Lee Sedol.

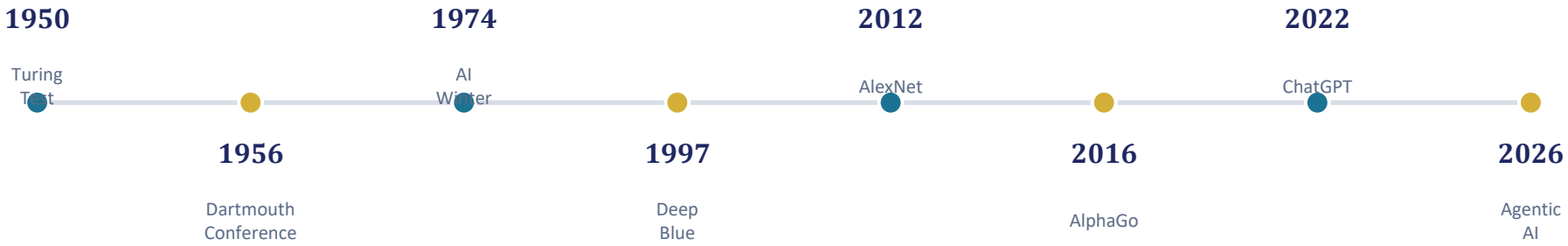
2022–Present

Generative AI revolution: ChatGPT, Gemini, Claude, Midjourney.

2025+

Multimodal and Agentic AI — systems that see, reason and act.

Timeline at a Glance



Notice the pattern: hype → unmet expectations → "winter" → breakthrough. Deep Learning's 2012 breakthrough finally broke this cycle thanks to data, compute and algorithms aligning together.

04

DL Successes — The Last Decade

Landmark breakthroughs that proved Deep Learning works at scale

Case Study — AlexNet & ImageNet (2012)

The Challenge

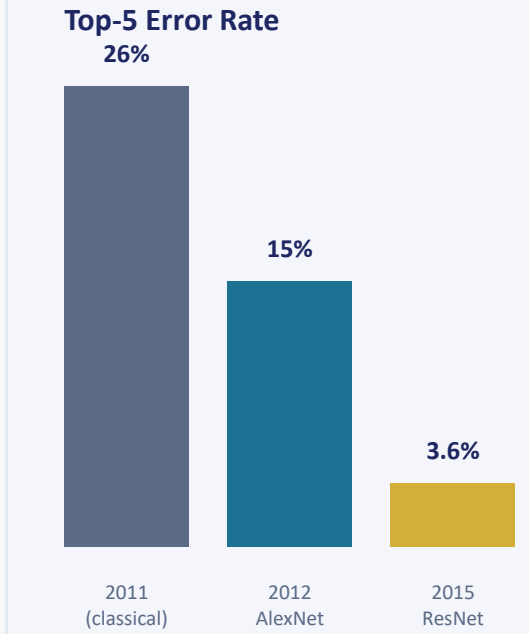
Classify 1.2 million images into 1000 categories — the annual ImageNet (ILSVRC) competition.

The Breakthrough

Krizhevsky, Sutskever & Hinton's AlexNet, a deep Convolutional Neural Network trained on GPUs, slashed the error rate from ~26% to ~15%.

The Impact

Proved that deep networks + GPUs + large data beat hand-engineered features — triggering the modern Deep Learning boom.



Case Studies — AlphaGo (2016) & ChatGPT (2022)

AlphaGo (DeepMind, 2016)

Go has more possible positions than atoms in the universe — brute force is impossible.

Combined deep neural networks with reinforcement learning and tree search.

Defeated 18-time world champion Lee Sedol 4–1, decades ahead of expert predictions.

ChatGPT & Generative AI (2022–)

Large Language Models, trained on internet-scale text with the Transformer architecture, learn to generate fluent, context-aware language.

Reached 100 million users within two months of launch — the fastest-adopted consumer application in history.

Sparked the wider Generative AI wave: Gemini, Claude, Midjourney, and beyond.

05

Reasons to Go Deep

Why now? Four forces that made Deep Learning practical

Four Forces Behind the Deep Learning Boom

1

Data

Explosion of labelled data — ImageNet, web text, sensors, smartphones — gives networks enough examples to learn from.

2

Compute

GPUs (and later TPUs) made the massive matrix multiplications of deep networks fast and affordable.

3

Algorithms

ReLU activations, Dropout, Batch Normalization, residual connections and the Transformer made very deep networks trainable.

4

Automatic Features

Networks learn their own features from raw data, removing the bottleneck of manual feature engineering.

CPU vs GPU vs TPU

Why parallel hardware is the engine behind deep learning

CPU vs GPU: Architecture

Two very different designs for two very different jobs

CPU (Central Processing Unit)

- 4–64 powerful cores, optimized for low-latency, sequential logic
- High clock speed (~3–5 GHz) per core
- Large per-core cache; complex branch prediction
- MIMD-style: each core runs a different instruction stream
- Best at: control flow, I/O, diverse general-purpose tasks

GPU (Graphics Processing Unit)

- Thousands of small cores (e.g., 10,000+ CUDA cores)
- Lower clock speed, but massive SIMT parallelism
- Very high memory bandwidth (~1–3 TB/s HBM)
- SIMD/SIMT: same instruction, thousands of data lanes at once
- Best at: throughput on repetitive, parallel math

Why deep learning needs GPUs

Training a neural network is dominated by matrix multiplications and convolutions — millions of independent multiply-accumulate operations per layer. GPUs run the same operation across thousands of cores simultaneously and use high-bandwidth memory to keep those cores fed, giving 10–100x speedups over CPUs. Tensor Cores further accelerate mixed-precision matmuls used in training.

06

Applications of Deep Learning

Where these techniques are changing industries today

Healthcare, Finance & Education

H

Healthcare

CNNs detect tumours, fractures and diabetic retinopathy from medical scans, often matching specialist accuracy.

Case study: a hospital receiving 50,000 X-rays/day uses DL to flag suspicious scans for radiologists — assisting, not replacing, doctors.

$

Banking & Finance

Deep models flag fraudulent transactions in real time by learning subtle, evolving patterns of misuse.

Case study: card networks use sequence models to score every transaction in milliseconds.

E

Education

Adaptive learning platforms personalise content and pace using a student's own performance data.

Case study: recommendation engines suggest the next topic or question based on mastery level.

Vision, Language & Everyday Life

V

Computer Vision

Self-driving cars use CNNs to detect lanes, pedestrians and obstacles from camera and LiDAR feeds.

Case study: Tesla / Waymo perception stacks run multiple deep networks per frame, 30+ times a second.

L

NLP & Chatbots

Transformer-based language models understand and generate human language for translation, search and assistants.

Case study: ChatGPT, Google Translate and voice assistants like Alexa and Siri.

★

Entertainment

Recommender systems learn viewing/listening patterns to personalise content feeds.

Case study: Netflix, Spotify and YouTube recommendation engines.

07

Challenges of Deep Learning

Deep Learning is powerful — but not free of serious limitations

Key Limitations to Be Aware Of

Data Hungry

Deep networks typically need thousands to millions of labelled examples; performance degrades sharply with small datasets.

Computationally Expensive

Training large models requires powerful GPUs/TPUs, significant time and substantial energy consumption.

Black-Box Nature

Predictions are hard to interpret — this is the core motivation for Explainable AI (XAI) techniques like Grad-CAM and SHAP.

Overfitting & Bias

Models can memorise training data or inherit and amplify biases present in it, hurting generalisation and fairness.

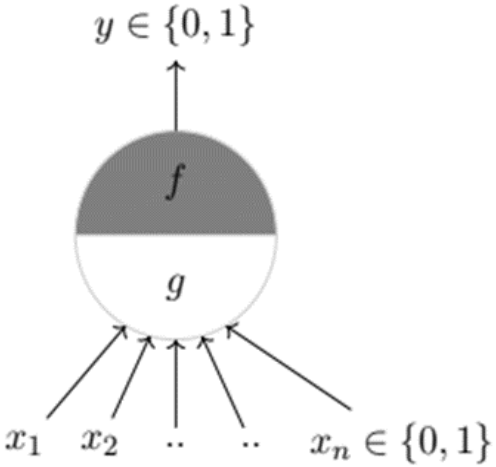
Adversarial Vulnerability

Small, imperceptible input perturbations can fool deep networks into confident, wrong predictions.

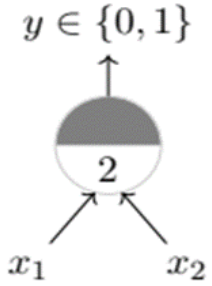
Lacks True Understanding

Deep Learning finds statistical patterns; it does not possess genuine reasoning or consciousness.

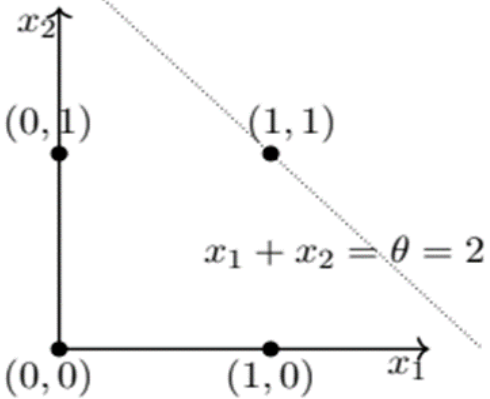
McCulloch Pit's Neuron (MP Neuron)



$$\hat{y} = 1 \; if \sum_{i=1}^{n} x_i \geq b$$
$$\hat{y} = 0 \; otherwise$$



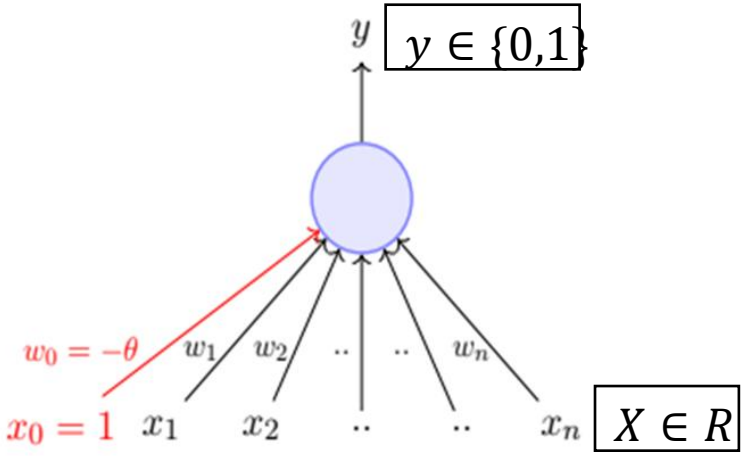
AND function
$x_1 + x_2 = \sum_{i=1}^{2} x_i \geq 2$



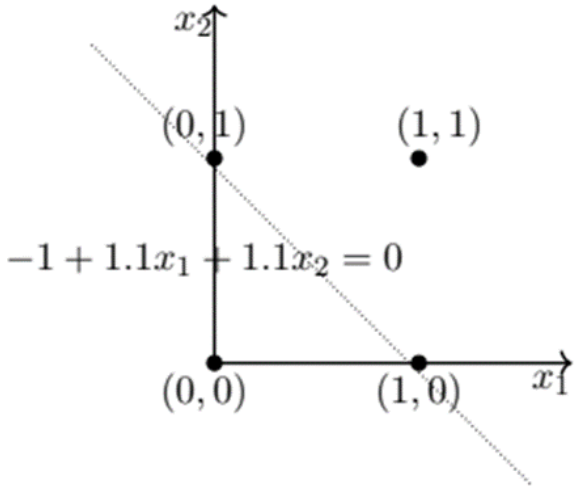
Six Jars for MP Neuron

	Data	Task	Model	Loss Function	Learning Algorithm	Evaluation metrics
MP Neuron	Boolean {0,1}	Binary Classification	$\hat{y} = 1 \; if \sum_{i=1}^{n} x_i \geq b$ $\hat{y} = 0 \; otherwise$	$Loss = \sum_{i=1}^{n} (y - \hat{y})^2$	Brute force	Accuracy = $\frac{correct\ predictions}{Total\ predictions}$

Perceptron



$$\hat{y} = 1 \; if \sum_{i=0}^{n} w_i x_i \geq 0$$
$$\hat{y} = 0 \; otherwise$$



while !*convergence* **do**
  Pick random $\mathbf{x} \in P \cup N$;
  if $\mathbf{x} \in P$ *and* $\sum_{i=0}^{n} w_i * x_i < 0$ **then**
    $\mathbf{w} = \mathbf{w} + \mathbf{x}$;
  end
  if $\mathbf{x} \in N$ *and* $\sum_{i=0}^{n} w_i * x_i \geq 0$ **then**
    $\mathbf{w} = \mathbf{w} - \mathbf{x}$;
  end
end

Unit 1 Summary

- Neural Networks are inspired by biological neurons; ANNs stack layers of these neurons to learn complex functions.
- Deep Learning automatically learns hierarchical features from raw data, removing the need for manual feature engineering.
- AI's history (1950–today) shows cycles of hype and “winters” — broken in 2012 by the convergence of data, compute and algorithms.
- Landmark successes — AlexNet, AlphaGo, ChatGPT — demonstrate Deep Learning's real-world impact.
- Deep Learning powers healthcare, finance, vision, language and entertainment applications — but remains data-hungry, costly and often a “black box.”

Thank You

Questions & Discussion | Next: Unit 2 — Perceptrons, Activation Functions & Training Deep Networks